

02

PERBEDAAN **UI**, **UX** and **USABILITY**

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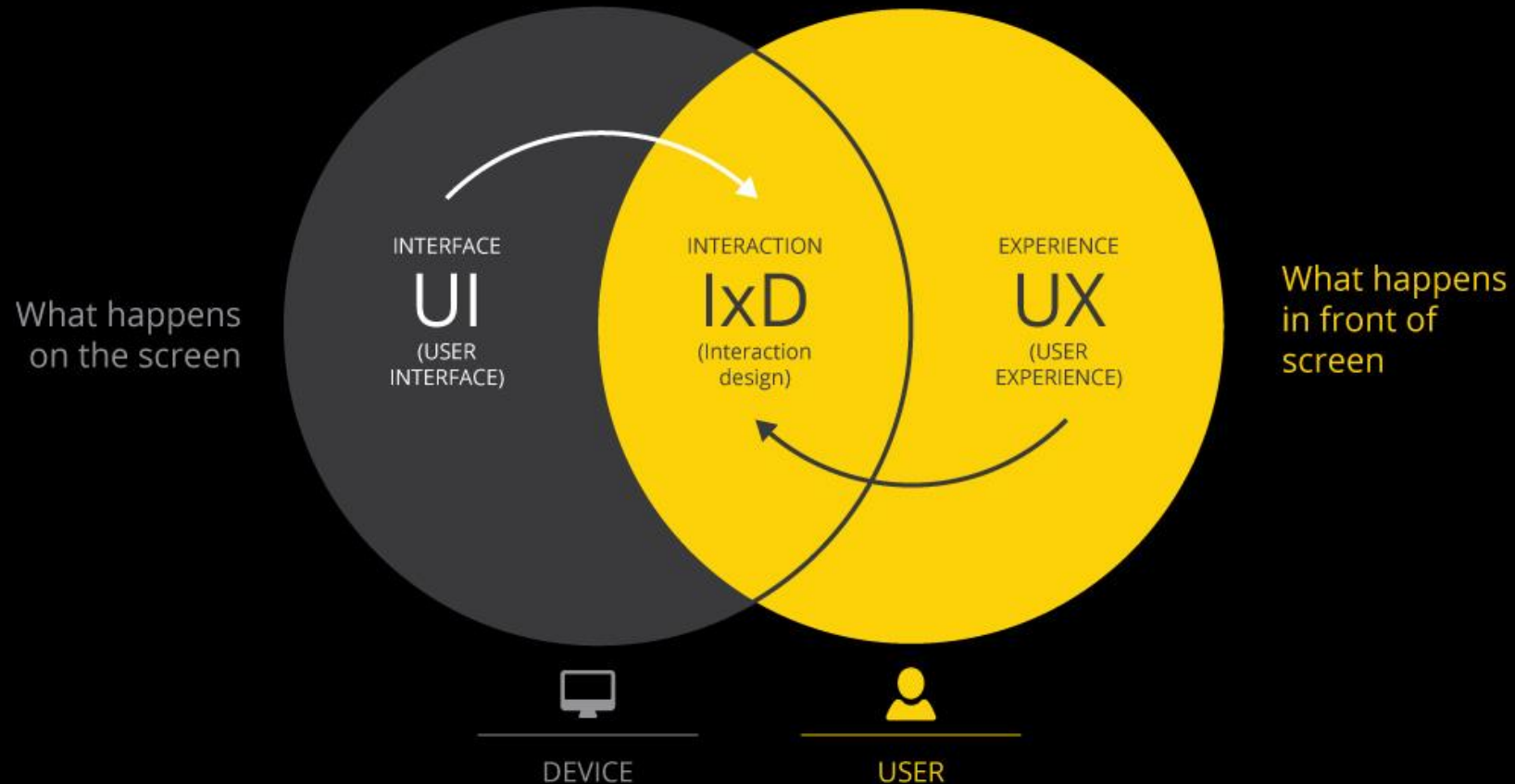
SISTEM INFORMASI
FAKULTAS ILMU KOMPUTER - UNIVERSITAS BRAWIJAYA



OUTLINE

- **Perbedaan UI dan UX**
- **User Interface**
- **Usability**
- **User Experience Honeycomb**
- **Usability Heuristic for UI Design**

UI vs IxD vs UX



USER INTERFACE (UI)

- The **space** where **interactions** between **humans** and **machines** occur ([Wikipedia](#))
- All **components** of an interactive system (software or hardware) that **provide information and controls** for the **user to accomplish specific tasks** with the interactive system ([ISO 9241-210](#))

USER INTERFACE (UI)

The **space** where **interactions** between **humans** and **machines** occur ([Wikipedia](#))

The UI includes hardware (physical) and software (logical) components.

UI provide a means of:

- **Input** – allowing the users to manipulate a system (i.e using it)
- **Output** – allowing the system to indicate the effects (give feedback) of the users' manipulation

USER INTERFACE (UI)

- UI **penting** agar user dapat merasakan UX.
- UI **berbeda** dengan UX
- UI yg menarik **belum tentu** memiliki UX yg baik
- UI yg baik dapat **meningkatkan** UX

USER EXPERIENCE (UX)

Person's **perceptions and responses** resulting from the **use and/or anticipated use** of a product, system or service (ISO 9241-210)

- **NOTE 1-** User experience includes all the users' **emotions, beliefs, preferences, perceptions, physical and psychological responses, behaviours and accomplishments** that occur before, during and after use

USER EXPERIENCE (UX)

- **NOTE 2** - User experience is a consequence of **brand image, presentation, functionality, system performance, interactive behaviour and assistive capabilities of the interactive system**, the user's internal and physical state resulting from prior experiences, attitudes, skills and personality, and the context of use.
- **NOTE 3 - Usability**, when interpreted from the perspective of the users' personal goals, can include the kind of perceptual and emotional aspects typically associated with user experience. Usability criteria can be used to assess aspects of user experience.

USER EXPERIENCE (UX)

UX encompasses all aspects of the end-**user's interaction with the company, its services, and its products** (Norman & Nielsen)

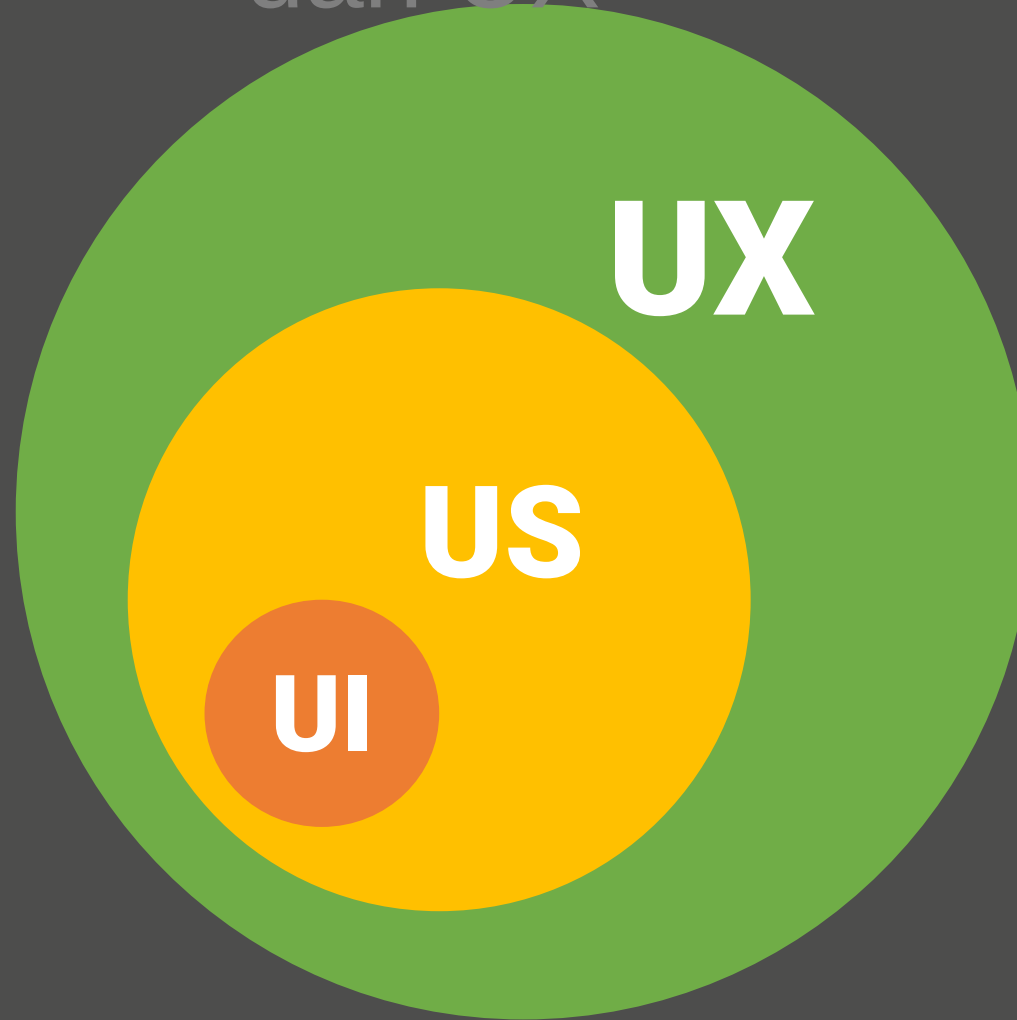
True user experience **goes far beyond** giving customers what they say they want, or providing checklist features.

In order **to achieve high-quality user experience** in a company's offerings there must be a seamless merging of the services of multiple disciplines, including engineering, marketing, graphical and industrial design, and interface design.

WHAT IS UX

The process of improving satisfaction by enhancing the usability and accessibility provided in the interaction between user and a product

HUBUNGAN UI, USABILITY dan UX



UX DESIGN SCOPE

DESIGN

- Interaction design
- Visual design
- Product design
- Feature writing
- Graphic arts
- Information Architecture
- Interaction design
- Interface layout

CONTENT

- Taxonomy creation
- Terminology creation
- Copywriting

RESEARCH

- Field research and interviewing
- Gathering and synthesizing data
- Documenting personas and findings
- Writing requirements
- Ideation coordination

TESTING

- Creation and administering of tests
- Usability and accessibility
- Prototyping

TEAM & CLIENTS

- Evangelizing, presenting & informing
- Working closely with programmers

USABILITY (Norman & Nielsen)

Usability is a **quality attribute** that assesses how easy user interfaces are to use
[\(Norman & Nielsen\)](#)

Utility = whether it **provides the features you need**.

Usability = how **easy & pleasant** these features are to use.

Useful = usability + utility

USABILITY (ISO-9241)

Usability - ISO 9241 definition

The **effectiveness**, **efficiency** and **satisfaction** with which specified users achieve specified goals in particular environments.

- **effectiveness**: the **accuracy and completeness** with which specified users can achieve specified goals in particular environments
- **efficiency**: the **resources expended** in relation to the accuracy and completeness of goals achieved
- **satisfaction**: the **comfort and acceptability** of the work system to its users and other people affected by its use

USABILITY (Norman & Nielsen)

Usability 5 main components [\(Norman & Nielsen\)](#)

- **Learnability:** How easy is it for users to accomplish **basic tasks** the **first time**?
- **Efficiency:** Once **users have learned** the design, **how quickly** can they perform tasks?
- **Memorability:** When **users return** to the design **after a period** of not using it, **how easily** can they reestablish proficiency?
- **Errors:** **How many errors** do users make, **how severe** are these errors, and **how easily** can they **recover** from the errors?
- **Satisfaction:** **How pleasant** is it to use the design?

Usability Requirement

- Effective
- Efficient
- Engaging
- Easy to learn
- Error tolerant

USABILITY (ISO-9241)

Effectiveness

- Accuracy → *Error rate*
- Completeness → *Successful task completeness*

Efficiency (resources expended)

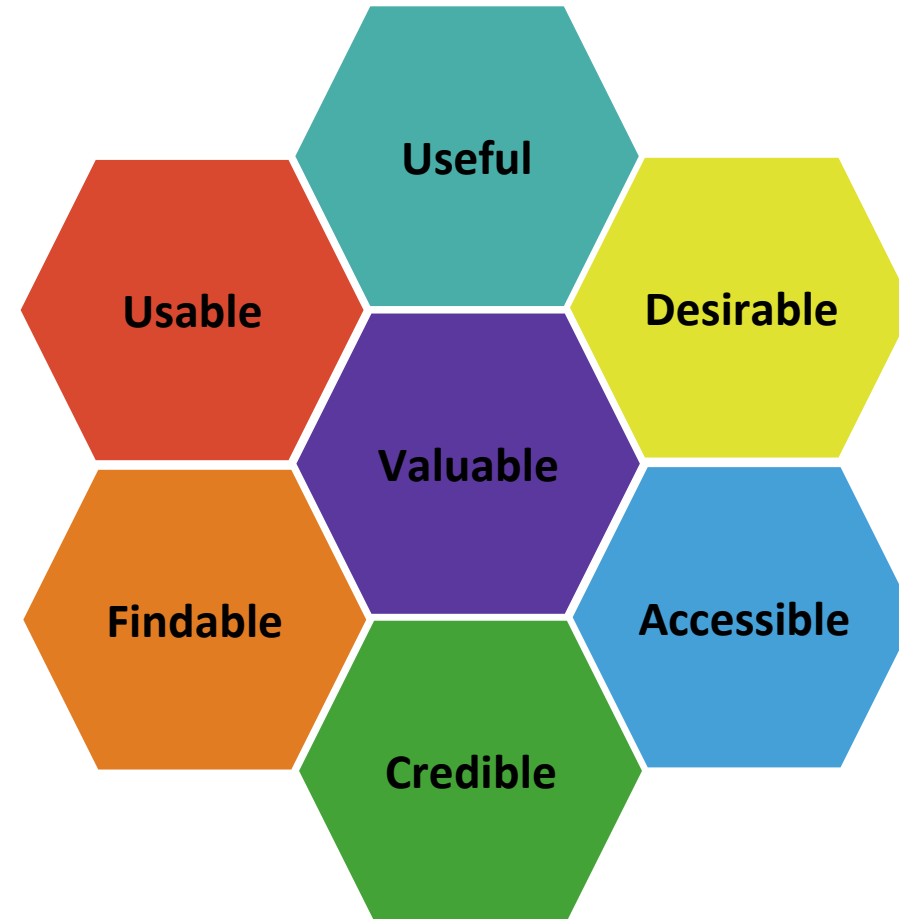
- Time → *Time on task*

Satisfaction (comfort and acceptability)

- *comfort and acceptability questionnaire*

UX Honeycomb

The UX honeycomb is a tool that helps to explain each **facet or quality of the user experience**.
([Peter Morville](#))



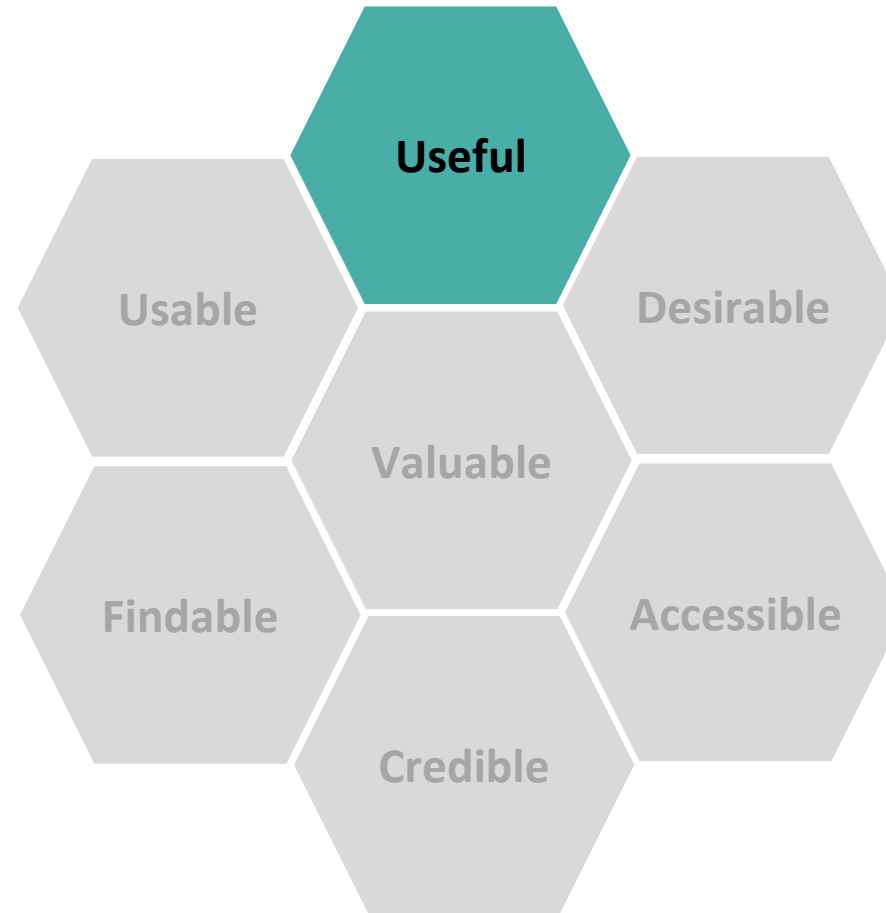
Read more:

http://semanticstudios.com/user_experience_design/

<https://medium.com/@danewesolko/peter-morvilles-user-experience-honeycomb-904c383b6886>

UX Honeycomb

Useful – A business's product or service **needs to be useful and fill a need**



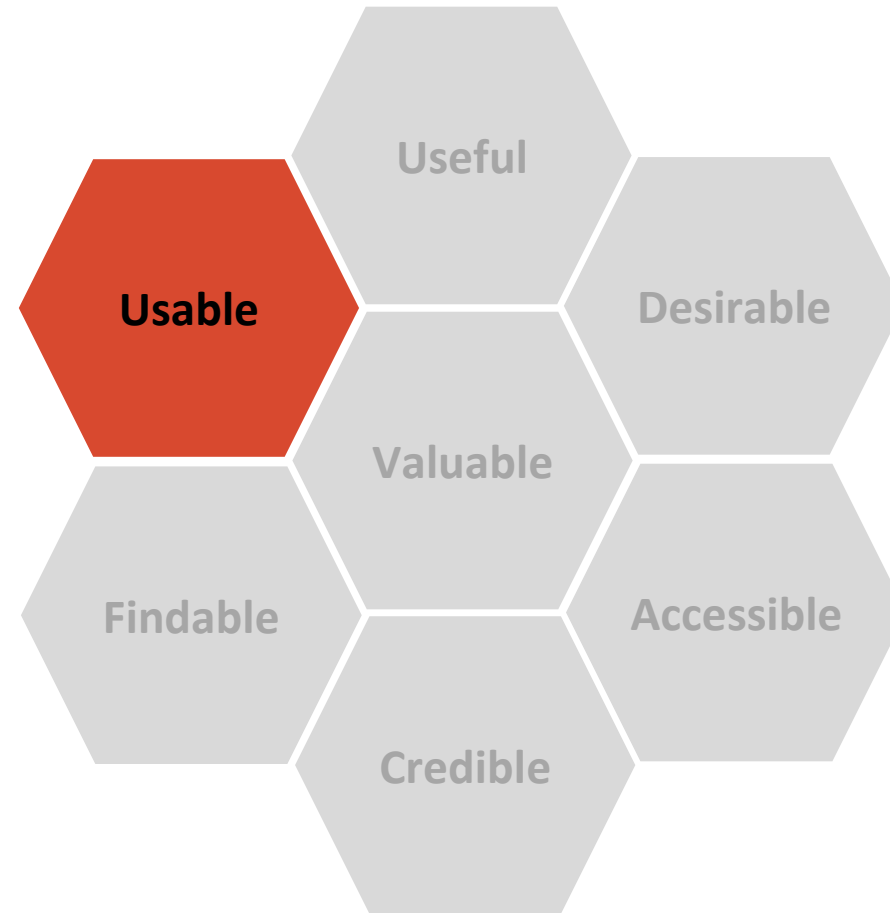
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UX Honeycomb

Usable – The system in which the product or service is delivered **needs to be simple and easy to use.**



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UX Honeycomb

Desireable – Image, identity, brand, and other design elements are used to **evoke emotion and appreciation**



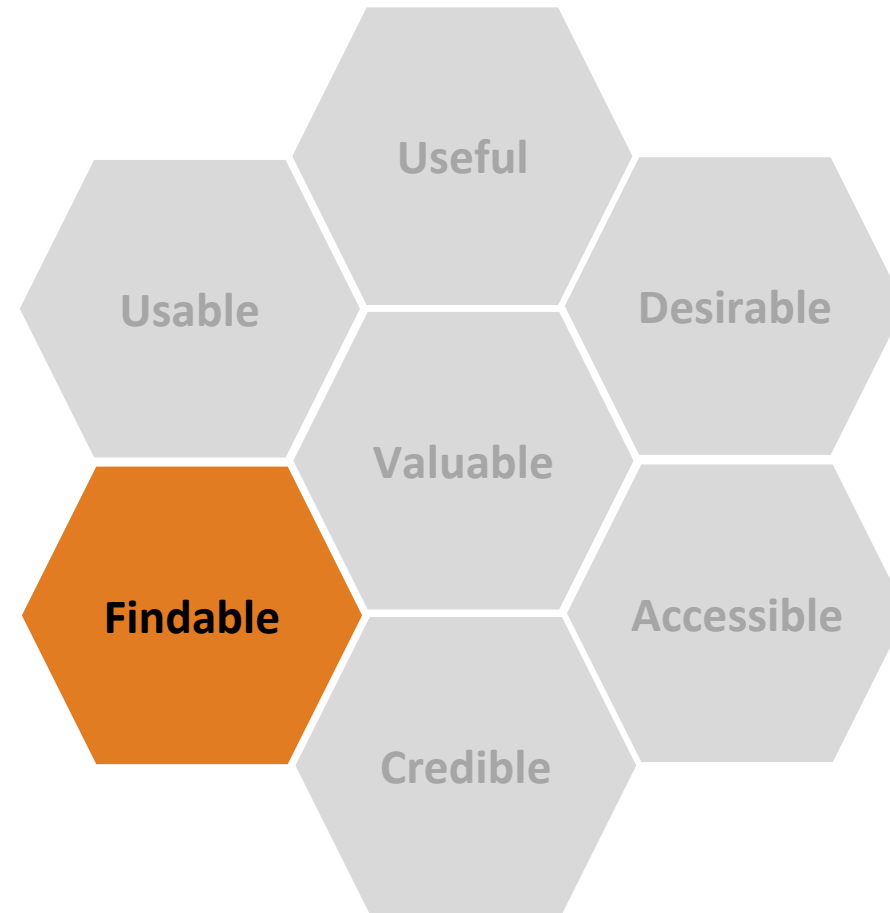
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UX Honeycomb

Findable – Information needs to be **findable and easy to navigate**.



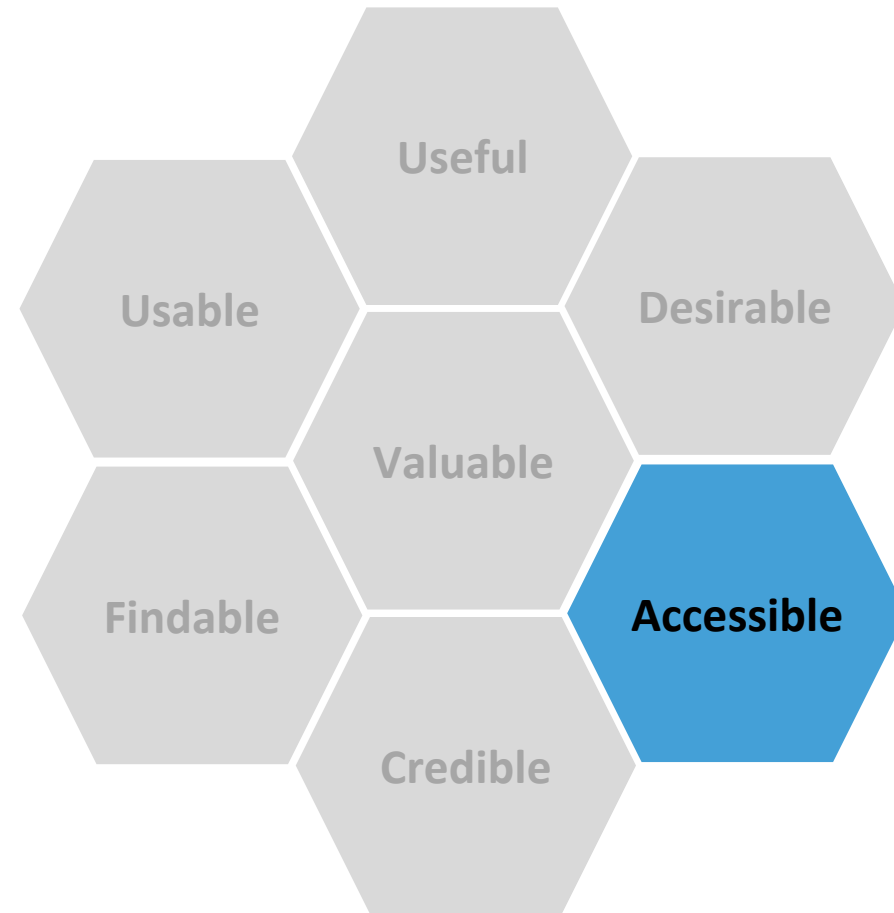
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UX Honeycomb

Accessible – The product or services should be designed so that even **users with disabilities** can have the same user experience as others.



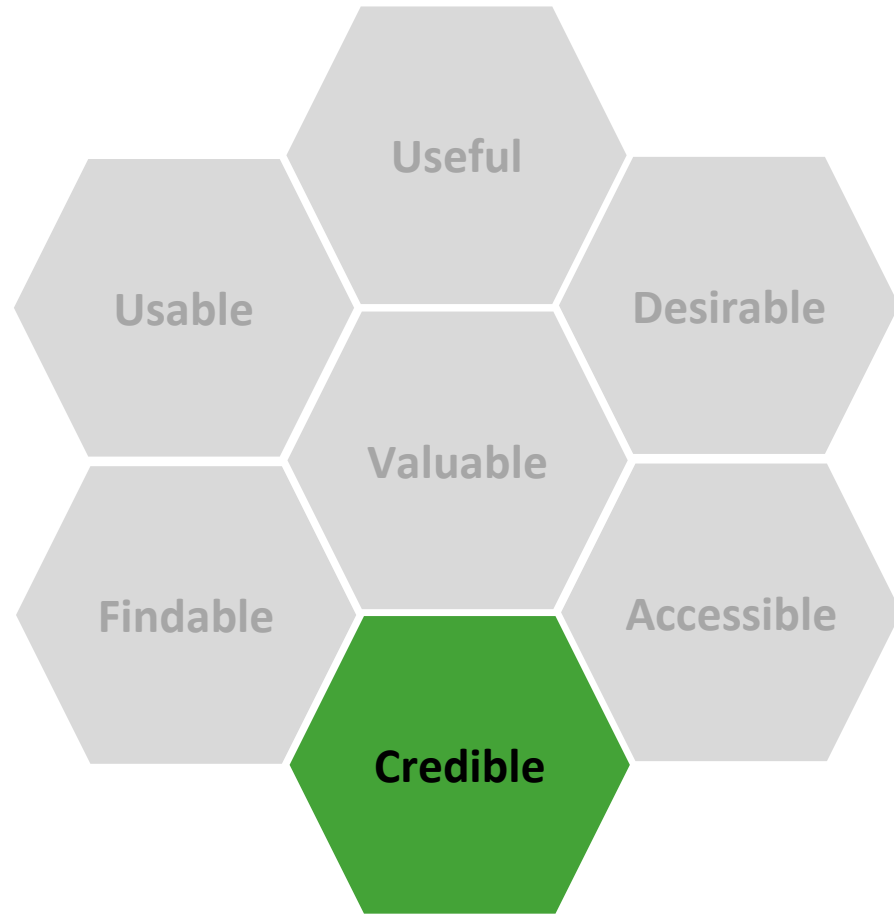
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UX Honeycomb

Credible – Influence users to **trust and believe** in what you tell them



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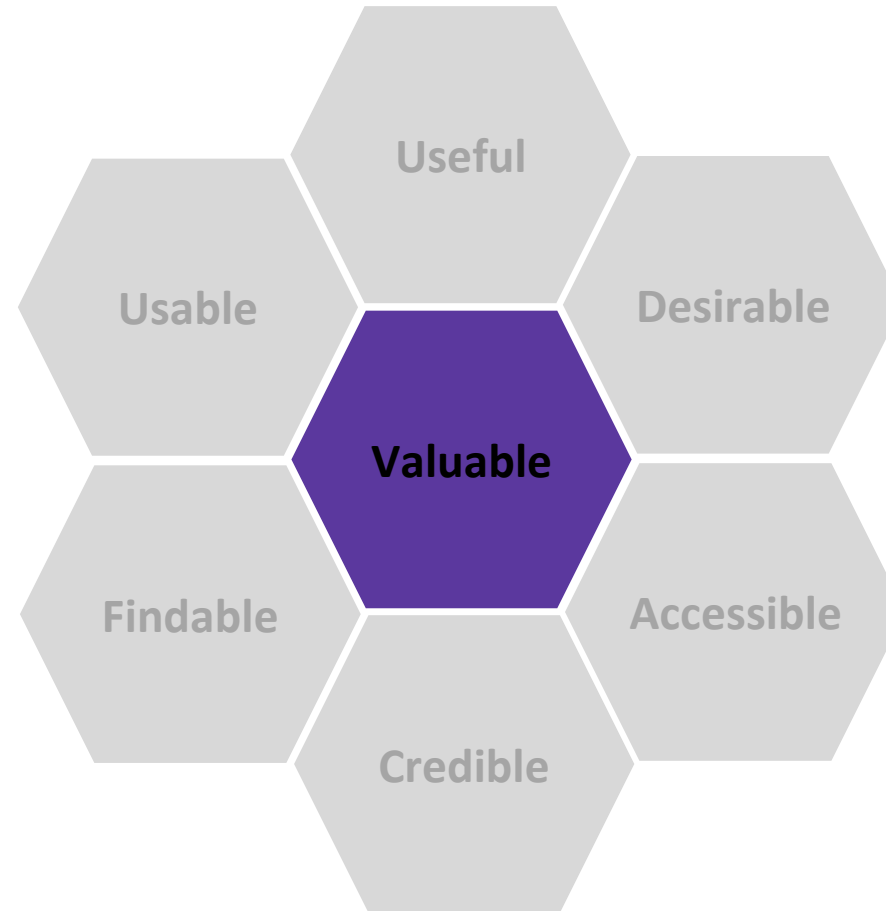
<https://medium.com/@danewesolko/peter-morvilles-user-experience-honeycomb-904c383b6886>

UX Honeycomb

Valueable – The product or services **must deliver value**.

For non-profits, the user experience must advance the mission.

With for-profits, it must contribute to the bottom line and improve customer satisfaction.



Read more:

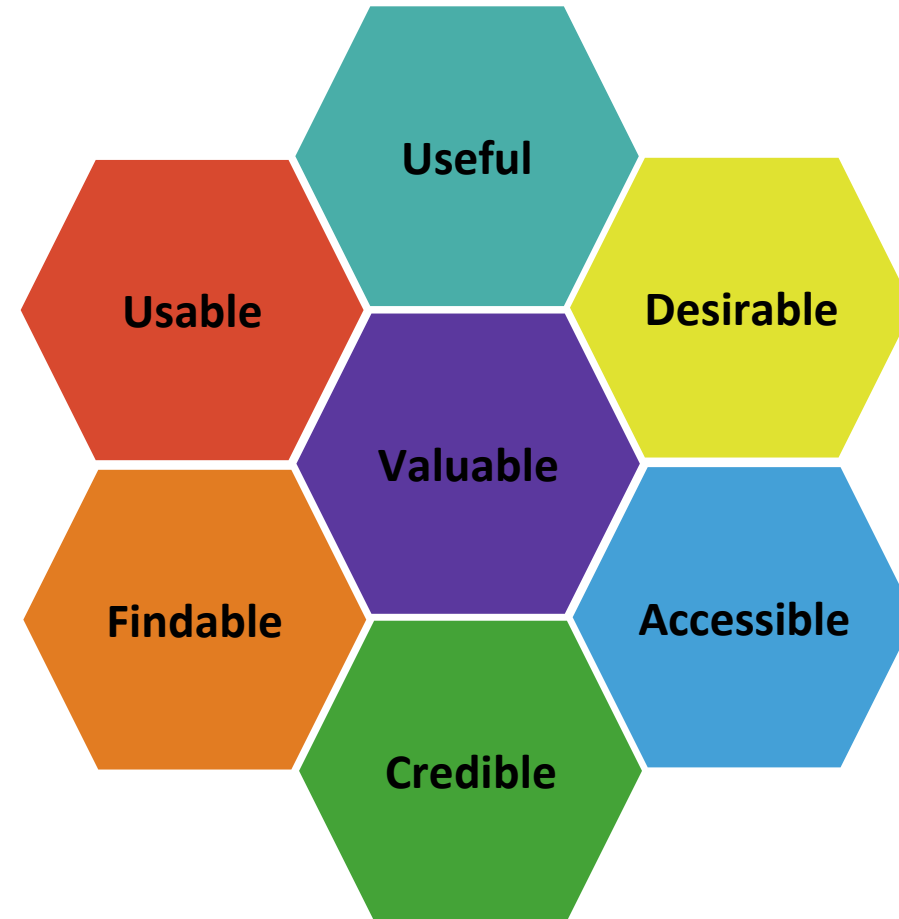
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UX Honeycomb

The UX honeycomb is a **helpful tool** for designers and stakeholders alike.

It helps to **understand** what and why UX is, **define priorities** in design, and serve as a tool for continuously **improving** areas of products and services.



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UI Principles

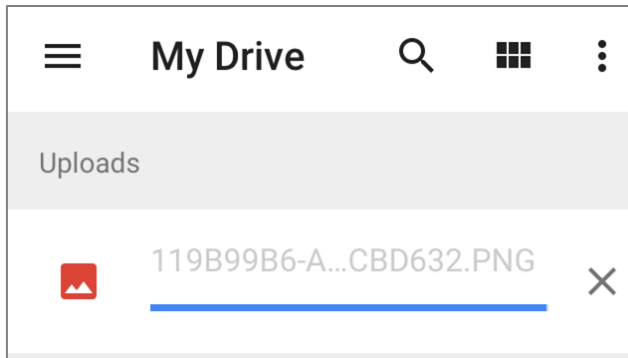
10 Usability Heuristics for UI Design ([Nielsen](#))

They are called "heuristics" because they are broad rules of thumb and not specific usability guidelines.

10 Usability Heuristics for UI Design

1. Visibility of system status

The system should always keep users **informed about what is going on**, through appropriate feedback within reasonable time.

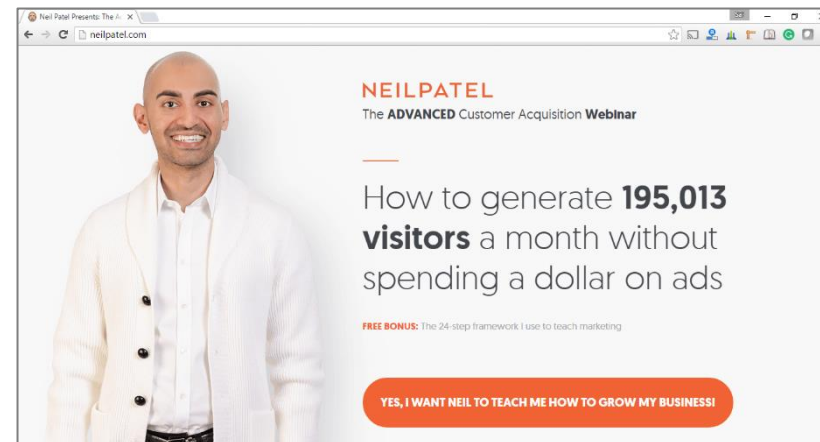


Sumber: <https://blog.prototypr.io/10-usability-heuristics-with-examples-4a81ada920c>

10 Usability Heuristics for UI Design

2. Match between system and the real world

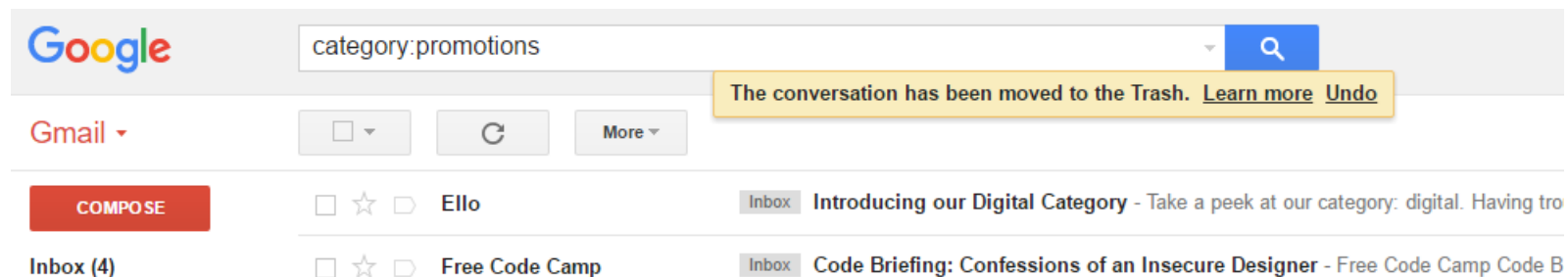
The system should **speak the users' language**, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.



10 Usability Heuristics for UI Design

3. User control and freedom

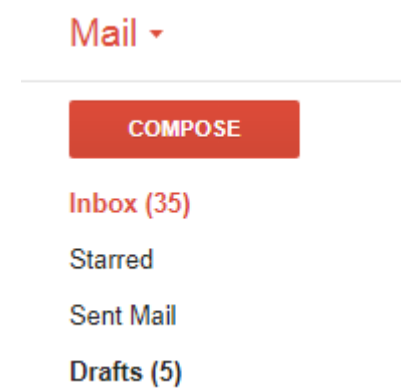
Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. **Support undo and redo and a clear way to navigate.**



10 Usability Heuristics for UI Design

4. Consistency and standards

Users **should not have to wonder** whether different words, situations, or actions mean the same thing. Follow platform conventions.



10 Usability Heuristics for UI Design

5. Error prevention

Even better than good error messages is a careful design which **prevents a problem from occurring** in the first place.

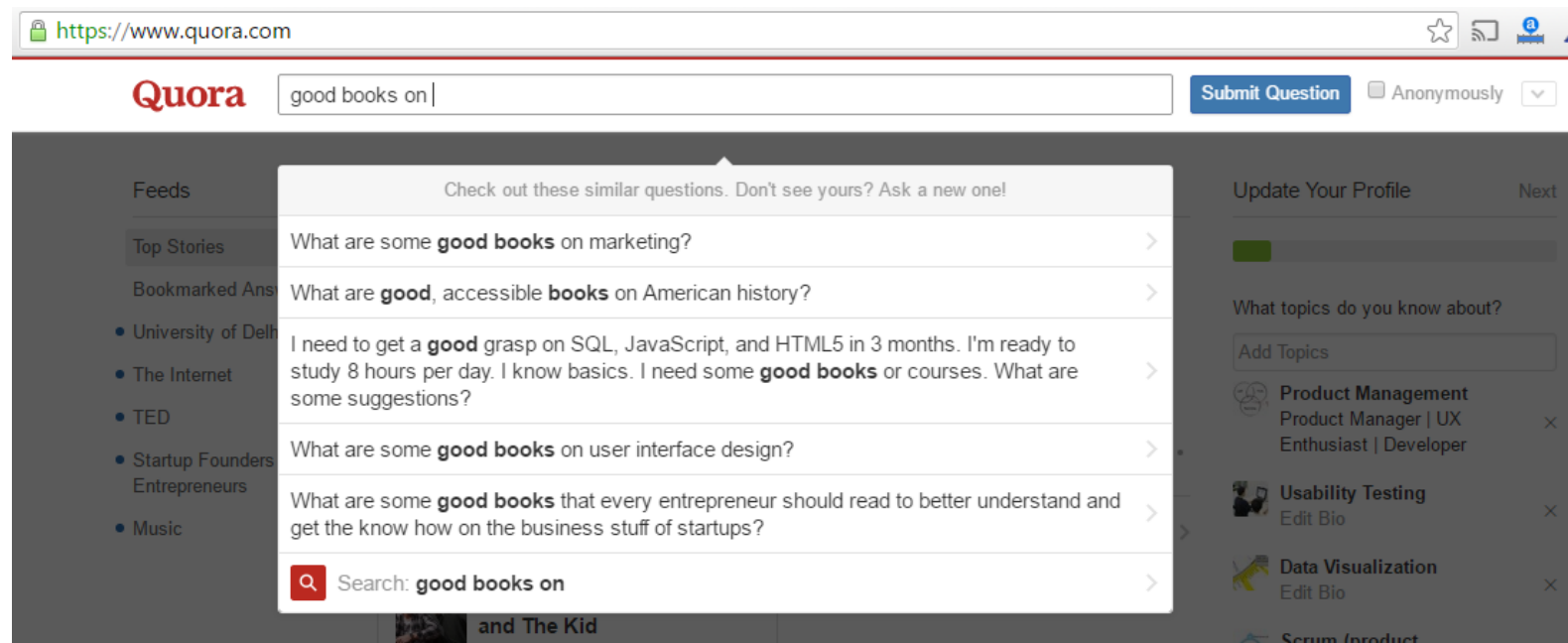
Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.



10 Usability Heuristics for UI Design

6. Recognition rather than recall

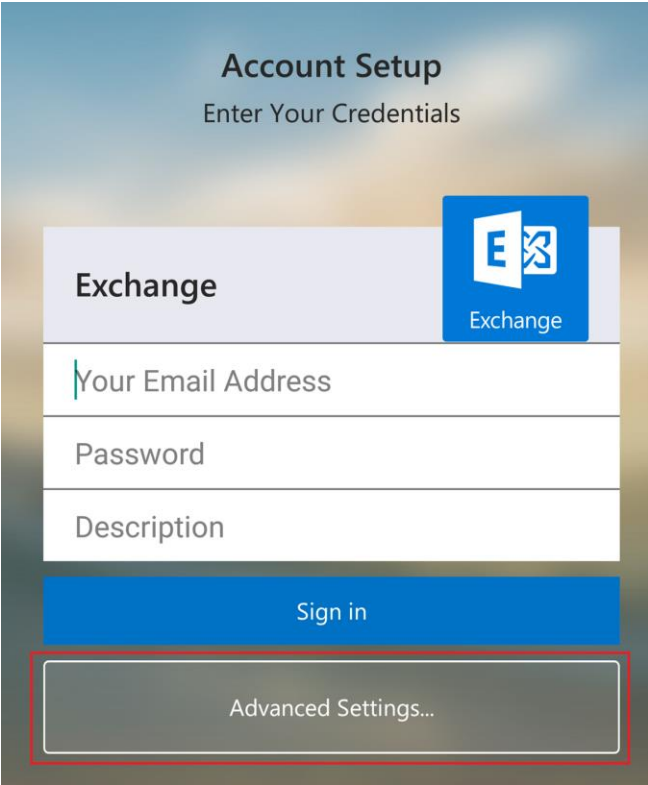
Minimize the user's memory load by making objects, actions, and options visible.



10 Usability Heuristics for UI Design

7. Flexibility and Efficiency of use

The Interface should be flexible **transforming** itself between a **novice user** and an **advanced user**.



The screenshot shows an 'Account Setup' window with the subtitle 'Enter Your Credentials'. It features a light blue header bar with the 'Exchange' logo and name. Below this, there are three input fields: 'Your Email Address', 'Password', and 'Description'. A prominent blue 'Sign in' button is located below the input fields. At the bottom, a grey button labeled 'Advanced Settings...' is highlighted with a red rectangular border, indicating an option for advanced users to customize the interface.

10 Usability Heuristics for UI Design

8. Aesthetic and minimalist design

Dialogues **should not** contain information which is **irrelevant or rarely needed**.



10 Usability Heuristics for UI Design

9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

The image displays two side-by-side login form mockups, each featuring a cartoon monkey logo at the top. Both forms have a red error message box at the top, indicating a failed login attempt. The left form shows an error message: "Sorry, we couldn't find an account with that username. Can we help you recover your username?" The right form shows an error message: "Sorry, that password isn't right. We can help you recover your password." Both forms include a "Username" field and a "Password" field, both with "I forgot" links. The "Password" field has a "Show" button. Below the fields are "Log In" and "Stay logged in" buttons. At the bottom, there are links for "Create an account" and "Trouble logging in?".

10 Usability Heuristics for UI Design

10. Help and documentation

It may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

GoDaddy Help



Domains



Linux Hosting (cPanel)



Windows Hosting (Plesk)



SSL Certificates



Office 365 from GoDaddy



Workspace Email



Website Builder



Account Management

Domain Related

[GoDaddy Auctions](#)

Online Marketing

[Get Found](#)

[Search Engine Visibility](#)

Partner Programs

[API Reseller](#)

[Basic & Pro Reseller](#)

Websites

[Design Services](#)

[InstantPage](#)

Referensi Heuristic Evaluation

- <https://www.nngroup.com/articles/ten-usability-heuristics/>
- <https://uxdesign.cc/user-experience-is-one-of-the-hottest-topics-in-day-today-designers-life-fb314978e1ff>

Berdasarkan pemahaman Anda,
jelaskan makna, ruang lingkup dan perbedaannya?



UX



US



UI

Lakukan evaluasi dari suatu aplikasi berdasarkan
10 Usability Heuristics for UI Design by Nielsen

